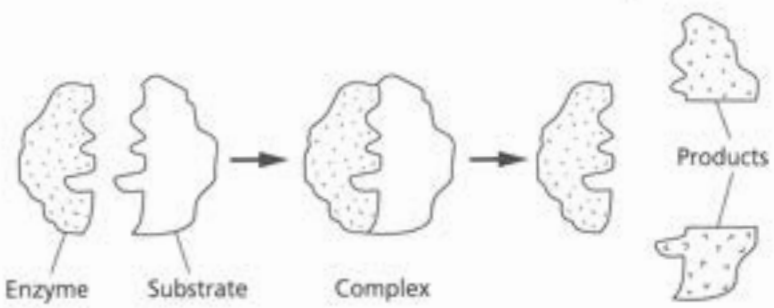
Homeostasis

Homeostasis is the maintenance of a balance between the functions carried out inside an organism and the environment, including body temperature and regulating levels of chemical compounds. Animals can be divided into *endotherms* or *ectotherms* depending on how their body temperature is regulated.

1 Outline the major differences in how the body temperatures of *endotherms* and *ectotherms* are regulated.

There are many factors that play a part in an animal's metabolism. Many complex reactions are occurring, both making large molecules and breaking them down. *Enzymes* (biological catalysts) are a vital part in ensuring that these reactions occur efficiently and at body temperature. Enzymes are very specific catalysts – most enzymes only catalyse a specific biochemical reaction. Enzymes are proteins that combine with a reacting molecule, the *substrate*. A simplified version of one model (*lock and key*) is shown in figure 1.

Figure 1



2 Copy figure 1 and use it to describe how the 'lock and key' model explains how enzymes work.

The shape of the enzyme might not be quite as fixed as suggested by this model. An *induced-fit* model may be more accurate. Some enzyme molecules may have their shape changed slightly by the substrate in order for the substrate to fit into the *active site* of the enzyme.

3 Using your diagram of figure 1, explain what is meant by the active site of the enzyme.

Reactions that give out heat are *exothermic* and the energy levels of products are lower than that of reactants. All reactions need *activation energy* to get them started. Catalysts lower the activation energy for reactions and enable them occur more quickly. Figure 2 shows the energy changes involved when a reaction is carried out both with and without a catalyst.

4 Copy figure 2. Identify points A, B, C and D on figure 2, choosing from: Activation energy (catalysed reaction), Activation energy (uncatalysed reaction), Energy, Energy released by reaction.

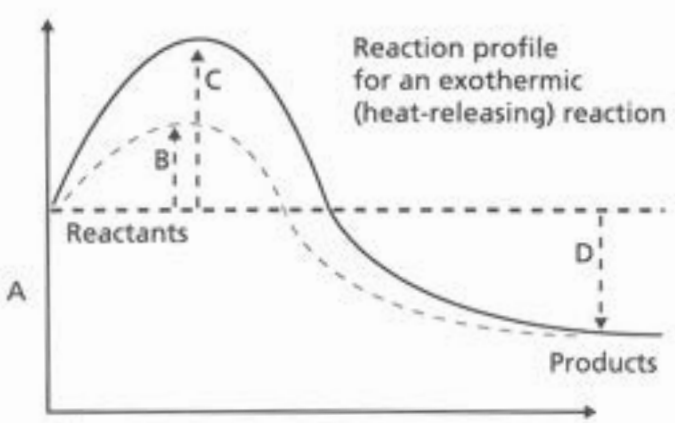


Figure 2

5 What does this define: '*A complex protein made by a living cell that catalyses a specific chemical reaction at a lower temperature than might normally be possible*'?

State whether the following statements are true or false:

6 All enzymes have the ending '-ase'. _____

7 Enzymes are usually small molecules. _____

8 Enzymes are specific in their actions. _____

9 In a reaction you start with products and end up with reactants. _____

10 Exothermic reactions take in heat. _____

11 Enzymes are proteins. _____

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Enzymes

Enzymes catalyse almost all of the reactions taking place in organisms and can be affected by:

- → temperature
- → pH
- → substrate

Temperature

Figure 3 shows how the rate of an enzyme-catalysed reaction varies with temperature.

12 Copy figure 3 and describe how the rate of the reaction shown alters with temperature.

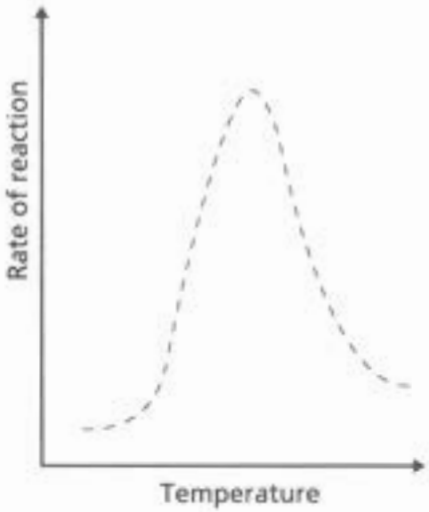


Figure 3

The *optimum temperature* for most enzymes is the temperature at which they are most efficient. In most mammals this is around body temperature. If the temperature rises much above 40°C, the enzymes are *denatured* – their structure is changed.

13 State the 'normal' internal temperature of a human.

pH

This is a logarithmic scale that indicates if a substance is an acid, base or neutral. Enzymes have an optimum pH range that varies from enzyme to enzyme. Enzyme activity decreases markedly more than one pH unit below or above optimum pH. Enzymes that work in the stomach will have an optimum pH of around 2, while those in human blood will have an optimum pH of just over 7.



Figure 4

14 A particular enzyme has an optimum pH of 6.5. Draw a labelled diagram (similar to figure 3) that shows how rate of reaction compares with pH for that enzyme.

Substrate concentration

Figure 5 shows how altering the amount of substrate present (keeping enzyme concentration constant) affects how quickly a particular reaction occurs.

15 Copy figure 5 and describe how the rate of the reaction shown alters with substrate concentration.

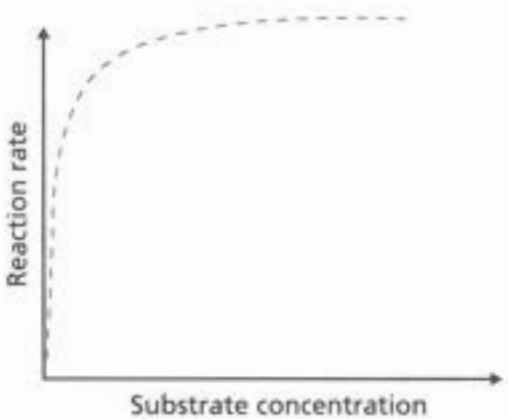


Figure 5

Identify the terms described by the following definitions:

16 *'The substance in a reaction acted on by an enzyme'.*

17 *'A number that indicates whether a substance is acidic, basic or neutral'.*

18 *'Process that changes structure of enzyme (eg by heating) so that active site no longer functions'.*

State whether the following statements are true or false:

19 Enzymes allow reactions to take place at moderate temperatures. _____

20 Enzymes are made in cells. _____

21 Breaking down large molecules is called 'synthesis'. _____

22 Enzymes involved in digestion in the stomach work best at pH values over 7. _____

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